

**Edward Casulli**  
Site Vice President

**Susquehanna Nuclear, LLC**  
769 Salem Boulevard  
Berwick, PA 18603  
Tel. 570.542.3795 Fax 570.542.1504  
Edward.Casulli@TalenEnergy.com



May 22, 2023

Attn: Document Control Desk  
U. S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION**  
**LICENSEE EVENT REPORT 50-387(388)/2023-001-00**  
**UNIT 1 LICENSE NO. NPF-14**  
**UNIT 2 LICENSE NO. NPF-22**  
**PLA-8068**

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**Docket No. 50-387**  
**and 50-388**

Attached is Licensee Event Report (LER) 50-387(388)/2023-001-00. The LER reports an event in which inadvertent contact between a relay cover and the relay trip contacts resulted in an automatic actuation of the 'B' Emergency Diesel Generator. This condition is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of any system listed in 10 CFR 50.73(a)(2)(iv)(B).

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "Ed Casulli".

E. Casulli

Attachment: LER 50-387(388)/2023-001-00

Copy: NRC Region I  
Mr. C. Highley, NRC Senior Resident Inspector  
Ms. A. Klett, NRC Project Manager  
Mr. M. Shields, PA DEP/BRP



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: [oir\\_submission@omb.eop.gov](mailto:oir_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

## 1. Facility Name

Susquehanna Steam Electric Station (SSES) Unit 2



050



052

## 2. Docket Number

00388

## 3. Page

1 OF 3

## 4. Title

Inadvertent Contact Between Relay Cover and Relay Trip Contacts Resulted in Automatic Actuation of the 'B' Emergency Diesel Generator

| 5. Event Date |     |      | 6. LER Number |                   |         | 7. Report Date |     |      | 8. Other Facilities Involved |                                         |               |
|---------------|-----|------|---------------|-------------------|---------|----------------|-----|------|------------------------------|-----------------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Rev No. | Month          | Day | Year | Facility Name                | <input checked="" type="checkbox"/> 050 | Docket Number |
| 03            | 23  | 2023 | 2023          | - 001 -           | 00      | 05             | 22  | 2023 | SSES Unit 1                  |                                         | 00387         |
|               |     |      |               |                   |         |                |     |      | Facility Name                | <input type="checkbox"/> 052            | Docket Number |

|                   |                 |
|-------------------|-----------------|
| 9. Operating Mode | 10. Power Level |
| 5                 | 000             |

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

| 10 CFR Part 20                              | <input type="checkbox"/> 20.2203(a)(2)(vi) | 10 CFR Part 50                              | <input type="checkbox"/> 50.73(a)(2)(ii)(A)            | <input type="checkbox"/> 50.73(a)(2)(viii)(A) | <input type="checkbox"/> 73.1200(a) |
|---------------------------------------------|--------------------------------------------|---------------------------------------------|--------------------------------------------------------|-----------------------------------------------|-------------------------------------|
| <input type="checkbox"/> 20.2201(b)         | <input type="checkbox"/> 20.2203(a)(3)(i)  | <input type="checkbox"/> 50.36(c)(1)(i)(A)  | <input type="checkbox"/> 50.73(a)(2)(ii)(B)            | <input type="checkbox"/> 50.73(a)(2)(viii)(B) | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)         | <input type="checkbox"/> 20.2203(a)(3)(ii) | <input type="checkbox"/> 50.36(c)(1)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(iii)              | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)      | <input type="checkbox"/> 20.2203(a)(4)     | <input type="checkbox"/> 50.36(c)(2)        | <input checked="" type="checkbox"/> 50.73(a)(2)(iv)(A) | <input type="checkbox"/> 50.73(a)(2)(x)       | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)   | 10 CFR Part 21                             | <input type="checkbox"/> 50.46(a)(3)(ii)    | <input type="checkbox"/> 50.73(a)(2)(v)(A)             | 10 CFR Part 73                                | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)  | <input type="checkbox"/> 21.2(c)           | <input type="checkbox"/> 50.69(g)           | <input type="checkbox"/> 50.73(a)(2)(v)(B)             | <input type="checkbox"/> 73.77(a)(1)          | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii) |                                            | <input type="checkbox"/> 50.73(a)(2)(i)(A)  | <input type="checkbox"/> 50.73(a)(2)(v)(C)             | <input type="checkbox"/> 73.77(a)(2)(i)       | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)  |                                            | <input type="checkbox"/> 50.73(a)(2)(i)(B)  | <input type="checkbox"/> 50.73(a)(2)(v)(D)             | <input type="checkbox"/> 73.77(a)(2)(ii)      | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)   |                                            | <input type="checkbox"/> 50.73(a)(2)(i)(C)  | <input type="checkbox"/> 50.73(a)(2)(vii)              |                                               |                                     |

☐ OTHER (Specify here, in abstract, or NRC 366A).

## 12. Licensee Contact for this LER

## Licensee Contact

Derek R. Smith, Senior Engineer – Nuclear Regulatory Affairs

## Phone Number (Include Area Code)

570-542-1377

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable To IRIS | Cause | System | Component | Manufacturer | Reportable To IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
|       |        |           |              |                    |       |        |           |              |                    |

## 14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

## 15. Expected Submission Date

| Month | Day | Year |
|-------|-----|------|
|       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 23, 2023, while reinstalling a relay cover on Engineered Safeguard System (ESS) Bus 2A202, inadvertent contact was made between the target reset lever on the relay cover and the relay trip contacts. This resulted in a trip of all 4.16 kV breakers on the 2B ESS Bus 2A202, an ESS bus lockout, and automatic start of the 'B' Emergency Diesel Generator (EDG). This condition is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of any system listed in 10 CFR 50.73(a)(2)(iv)(B). This event was also reported via Event Notification 56430 in accordance with 10 CFR 50.72(b)(3)(iv)(A).

The cause of the event was a momentary loss of focus on the relay target reset lever during reinstallation of the relay cover. This cause was attributed to a human performance error. Additional investigation concluded the potential impacts of an inadvertent error associated with the relay reinstallation activities were not adequately mitigated. Corrective actions included performing accountability per station procedures and planned procedure revisions to include continuous use checklists identifying removal and restoration work steps associated with relays that have the potential to trip protective devices.

There were no actual consequences to the health and safety of the public.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
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|                                                                                    |                                                |                                         |                      |                                 |                  |
|------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|----------------------|---------------------------------|------------------|
| <b>1. FACILITY NAME</b><br><br>Susquehanna Steam Electric Station (SSES)<br>Unit 2 | <input checked="" type="checkbox"/> <b>050</b> | <b>2. DOCKET NUMBER</b><br><br>05000388 | <b>3. LER NUMBER</b> |                                 |                  |
|                                                                                    | <input type="checkbox"/> <b>052</b>            |                                         | YEAR<br>2023         | SEQUENTIAL<br>NUMBER<br>- 001 - | REV<br>NO.<br>00 |

**NARRATIVE****CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power (RTP)

Unit 2 – Mode 5, zero percent RTP

The described activity was performed during the Division 2 electrical maintenance outage with Division 1 systems protected. There were no structures, systems, or components that were inoperable at the start of the event and contributed to the event.

**EVENT DESCRIPTION**

On March 23, 2023, while reinstalling a relay cover on Engineered Safeguard System (ESS) [EIS System Code: B] Bus 2A202 [EIS System/Component Code: E/BU], inadvertent contact was made between the target reset lever on the relay [EIS Component Code: RLY] cover and the relay trip contacts. This resulted in a trip of all 4.16 kV breakers [EIS Component Code: BKR] on the 2B ESS Bus 2A202, an ESS bus lockout, and automatic start of the 'B' Emergency Diesel Generator (EDG) [EIS System/Component Code: EK/DG]. The 'B' EDG operated as expected.

The following timeline demonstrates the sequence of events:

03/23/2023 ~ 15:45: Relay and Test lead technician conducted a pre-job brief for critical relay calibrations on ESS Bus 2A202.

03/23/2023 ~ 16:15 – 17:00: Technicians arrived at the work area and performed a two minute drill. Removal, calibration, and reinstallation of the A-phase bus differential relay was completed.

03/23/2023 ~ 17:00 – 17:30: Removal and calibration activities for the B-phase bus differential relay was completed.

03/23/2023 ~ 17:33: During installation of the relay cover, inadvertent contact was made between the target reset lever on the relay cover and the relay contacts. This resulted in a trip of the 2B Bus lockout relays which opened all the bus supply breakers and started the 'B' EDG.

3/23/23 ~ 17:37: All work was stopped and the Unit 2 control room was contacted.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of any system listed in 10 CFR 50.73(a)(2)(iv)(B). This event was also reported via Event Notification 56430 in accordance with 10 CFR 50.72(b)(3)(iv)(A).

**CAUSE OF EVENT**

The cause of the event was a momentary loss of focus on the relay target reset lever during reinstallation of the relay cover. This cause was attributed to a human performance error. Additional investigation concluded the potential impacts of an inadvertent error associated with the relay reinstallation activities were not adequately mitigated.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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|                                                                                    |                                                |                                         |                      |                                 |                  |
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|                                                                                    | <input type="checkbox"/> <b>052</b>            |                                         | YEAR<br>2023         | SEQUENTIAL<br>NUMBER<br>- 001 - | REV<br>NO.<br>00 |

**NARRATIVE****ANALYSIS/SAFETY SIGNIFICANCE**

Based on engineering analysis of the event, with the failure of one on-site circuit (i.e., 'B' EDG), the three remaining buses could achieve safe shutdown of the operating unit during all design basis accidents, consistent with the design and licensing basis of the plant. There was sufficient redundancy in systems impacted by the failure such that their safety function would still be performed. Safe shutdown was also maintained for Unit 2, which was in Mode 5 for the refueling and inspection outage. Therefore, the electrical power system could have fulfilled its safety function for the condition described.

Accordingly, this event will not be counted as a safety system functional failure in the Reactor Oversight Process Performance Indicator. There were no actual consequences to the health and safety of the public as a result of this event.

**CORRECTIVE ACTIONS**

Corrective actions included performing accountability per station procedures and planned procedure revisions to include continuous use checklists identifying removal and restoration work steps associated with relays that have the potential to trip protective devices.

**COMPONENT FAILURE INFORMATION**

Not applicable.

**PREVIOUS OCCURRENCES**

None.